Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/08/2016
PI Name:	Blemker, Silvia Ph.D.		
Project Title:	Development of a Muscle Adaptation in Space-flight Simulator (MASS)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22903-3390	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	09/01/2016	**End Date:**	09/30/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project closed out early as of 9/30/2017; original end date was 8/31/2019, per B. Lewandowski/JSC and NSSC information (Ed., 3/19/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Peirce-Cottle, Shayn Ph.D. (University of Virginia) Fregly, Benjamin Ph.D. (University of Florida)		
Grant/Contract No.:	NNX16AR98G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Muscle loss due to reduced loading in microgravity remains a major limiting issue for long-term spaceflight. Though exercise countermeasures have been developed and implemented on the Intenational Space Station (ISS), quantitative methods for predicting how skeletal muscle adapts structurally and functionally over time in response to exercise in a microgravity environment do not yet exist. A computational modeling approach that can predict changes in muscle properties as a function of time in microgravity and a predetermined amount of in-flight exercise stimulus will facilitate the design of exercise programs that can be employed both pre-flight and in-flight to effectively combat muscle atrophy and enhance muscle strength both during and after flight. This proposal seeks to develop a multiscale computational model that can predict changes in muscle structure and function in response to both microgravity and in-flight exercise. Our modeling approach will unite -- into a cohesive, user-friendly, and open-source framework -- computational simulations at different levels of spatial scale, including: dynamic OpenSim simulations at the whole body level, finite-element simulations of muscle tissue, and agent-based simulations of muscle cells that include the biochemical and biomechanical processes that drive individual muscle cell adaptations. This level of integration is necessary because, though they are often only considered in isolation, factors at all of these scales influence the degree of muscle atrophy during prolonged spaceflight. Our team brings together highly unique expertise: Silvia Blemker (modeling of muscle mechanics/physiology), B.J. Fregly (dynamic modeling of movement), and Shayn Peirce-Cottler (agent-based modeling and cellular/tissue adaptation). Our proposed model will integrate with OpenSim biomechanics simulation software and will be quantitatively verified and validated following the standard of NASA-STD-7009. Documentation that describes the modeling approaches used, as well as a model operations guide, will be provided as deliverables. The software source code will also be provided to NASA as an open-source product.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2016.
Bibliography Type:	Description: (Last Updated:)